

Jayana Nanayakkara

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.A. in Computer Science

May 2027

Coursework: CS 61B (Data Structures & Algorithms), CS 61A (Interpretation of Programs), Data 8 (Data Science), Machine Learning, Object Oriented Programming (Java), Algorithm Design (C++), Database Programming (SQL)

WORK EXPERIENCE

Software Engineer

February 2026 – Present

Berkeleytime

Berkeley, CA

- Built private **MongoDB** backup system using **Kubernetes CronJobs**, adding Cloudflare Access authentication.
- Implemented **GraphQL** endpoints to manage private backup access for staff members using **Cloudflare API**.
- Created onboarding **bash script** to set up Berkeleytime locally, reducing setup friction for new team members.

Software Engineer Intern

September 2025 – December 2025

Nelumbium Capital

Remote

- Built **Python Flask REST API** for financial risk analysis platform processing 4,000+ financial events daily.
- Developed data processing pipeline using **LangChain** & Nemotron models to transform CSVs into RDF triples.
- Stored data **AllegroGraph** database with optimized **SPARQL** queries achieving **sub-500ms** response times.
- Leveraged **NetworkX** for centrality analysis and path-finding algorithms to detect risk propagation patterns.

Software Engineer Intern

May 2024 – Aug 2024

Intel Corp: Graphics Numerical Hardware & System-Level Design Group

Folsom, CA

- Optimized arithmetic circuit software for integer multiplication, improving computational efficiency by **40%**.
- Automated time-intensive constraint-model testing with **Python** and the **MiniZinc API**, using **thread-based concurrency** to run solver instances in parallel and reduce runtime from **days to hours**.

Software Engineer Intern

Jan 2024 – May 2024

Personal Insurance Federation of California

Sacramento, CA

- Built **Spring Boot (Java)** service to process **100,000 documents** with **asynchronous** ingestion jobs.
- Created **text extraction** (AWS Textract) enabling **keyword/semantic search** (ElasticSearch) over documents.
- Developed a **RAG** workflow (**OpenAI API**) to answer queries about documents with citations.
- Collaborated with legal and legislative teams to develop user interface using **React TypeScript**.

Teaching Assistant

Aug. 2024 – Dec. 2024

CISD 307: Intro to Machine Learning @ Folsom Lake College

Folsom, CA

- Coordinated with a team of assistants to tutor students through concepts ranging from decision trees to **deep neural networks (PyTorch)** and scored/provided feedback on student code.
- Developed new curriculum such as recorded presentations and **Jupyter Notebooks** to help communicate advanced machine learning algorithms to students.

PROJECTS & EXTRACURRICULARS

Wealth Wise: Financial Education | Congressional App Challenge Winner

Aug 2023 – Present

- Designed and programmed frontend with investing lessons and chatbot UI using **React JavaScript** and **Next.js**.
- Developed a **dockerized Flask REST API** for user authentication and data persistence with JWT and Firebase.
- Deployed the containerized app to **AWS ECS (Fargate)**, implemented **CI/CD pipelines** with **GitHub Actions** for automated workflows.
- Won 1st** at the Congressional App Challenge, a nation-wide hackathon, reaching **10,000+ users**.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, Swift, SQL, MiniZinc

Frameworks & Libraries: Spring Boot, Flask, FastAPI, PyTorch, React, Kubernetes, TensorFlow, Prometheus

Technologies: AWS (ECS, RDS, S3), Docker, Git, PostgreSQL, Firebase (NoSQL), Linux, CI/CD